

Reinstatement out of equilibrium: shock speed and the destination of new work

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Abstract

Storck (2026) closes the spatial task-based model as a static equilibrium and reads its consequences as comparative statics between two rest points. This paper turns the same model into laws of motion and asks what the static account cannot: how the transition unfolds in time, and how its outcome depends on the speed of the shock. The technology matures gradually — a logistic diffusion over a calendar window — so that displacement, and the new tasks it seeds on the boundary of the technology field, are released as a flow rather than at once. The single new object the dynamics require is a stock of *unbound* task mass: the seeded work that has been created but not yet attached to any occupation, the third fate of new tasks that the static model creates and discards at every evaluation. Its trajectory $U(t)$ is the transitional mismatch between destruction and creation, and it is the model's central dynamic observable. Binding draws this stock down: new work is attracted to the occupations whose held capabilities best match it, but each occupation can absorb only at a rate limited by its size, so a saturated best match leaves a residue that cascades to the next-best match. The outcome is governed by a single ratio, the tempo of the shock against the tempo of redistribution. When redistribution keeps pace, reinstated work is bound by the high-skill specialists whose capabilities fit the frontier; when the shock outruns redistribution, small best-matched occupations cannot absorb it and the mass is taken instead by the large, lower-skilled occupations that can, a forced downgrading. The same technology and the same field thus yield materially different distributions of work depending only on the tempo: the final allocations of the slow and fast regimes correlate at 0.37. The two regimes are the proved limits of the binding law, and the destination moves monotonically between them with the tempo. This path-dependence of the destination, and not only of the transitional cost, is invisible to the equilibrium framework, whose fixed point sees only the endpoint. The realisation is illustrative, inheriting the calibration of Storck (2026).

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1 Introduction

A technology does not arrive all at once. It emerges, diffuses, and matures over years, and while it does, the labour market is already responding: displacement, the creation of new tasks, and the reallocation of workers run in parallel, not in sequence. The task-based framework cannot see this. Where it has dynamics they belong to the technology — which tasks are automated, when new ones arrive (Acemoglu and Restrepo 2018) — while the allocation of workers to tasks clears instantaneously, so the consequences of a technology are read as the difference between two rest points (Acemoglu and Restrepo 2019). The intermediate path of the allocation, and with it the cost of the transition and any dependence of the outcome on the speed of the shock, is by construction absent. Yet the transition is where the human consequences of automation are felt, and the central claim of this paper is that the speed of the shock relative to the speed of adjustment determines not merely how costly the transition is but *who ends up doing the reinstated work*.

The framework we build on is developed in two papers. Storck and Andersson (2026) construct the task space: tasks and occupations are mapped from O*NET statements onto a two-dimensional disk, occupations as bundles of tasks. Storck (2026) builds the model of technological change on that geometry: the market price of skill is a directional field over the space; a technology is a Gaussian augmentation field; automation removes the high-priced tasks of a bundle; and reinstatement seeds new tasks on the gradient boundary of the technology field, which survive as human work only where labour remains the cheaper producer. That model is closed as a static equilibrium, with the adjustment dynamics left explicitly for later work; the rise-and-fall of employment which Bessen (2019) documents over time appears in it only as a cross-section over the demand elasticity, not as a trajectory. This paper is that later work.

We make one structural addition and otherwise reuse the static model unchanged. That model creates new task mass on the boundary ring and partitions it into three fates — captured by capital, bound to an occupation, and unbound — but it has no home for the unbound part: it is created and discarded at every evaluation. The dynamic model gives it a home, a stock $U(\mathbf{r}, t)$ of unbound task mass that fills as the technology matures and drains as occupations bind it. This stock is the only genuinely new object; population, price, displacement, seeding, and the bundle operator are the static model's, now read as rates rather than as a solved equilibrium. The trajectory of the stock, $U(t)$, is the transitional mismatch between destruction and creation, and it is the first observable the static model cannot produce.

The second, and the paper's central result, concerns binding. Unbound work is attracted to the occupations whose held capabilities best match it, but an occupation can take on new work only at a rate limited by its size, so the best match, if small, saturates and leaves

a residue that is offered to the next-best match. The realised allocation therefore depends on a race: the tempo at which the shock seeds new work against the tempo at which occupations can bind it. We show that this ratio governs not only the height and duration of the transitional mismatch $U(t)$ but the final destination of the work. A gradual shock — redistribution faster than the shock — lets the best-matched occupations keep pace, and the reinstated work is bound by the high-skill specialists whose capabilities fit the frontier. A shock that outruns redistribution forces the work onto the large, lower-skilled occupations that can absorb it fastest, a downgrading that the gradual transition avoids. The two regimes are run through the same technology and the same field; their final allocations of reinstated work correlate at 0.37. Both endpoints are proved as limits of the binding law — absorption faster than the shock converges to the match-weighted allocation, absorption slower than the shock to initial size shares — and the realised destination moves monotonically between them as the tempo varies (Section 4.3). The destination, not just the transitional cost, is path-dependent, and the equilibrium framework sees none of it.

That reallocation takes time and shapes equilibrium is an old observation. In the islands tradition workers move slowly between markets and unemployment is the reallocation in progress (Lucas and Prescott 1974; Alvarez and Shimer 2011); distance appears as a cost, but there is no space in which the shock and the workers’ capabilities are located together. Şahin et al. (2014) measure the unemployment produced when job seekers search in the wrong markets: workers waiting for work. The unbound stock of this paper is the reverse object, work that has been created but not yet attached to any occupation — mismatch upstream of any vacancy — and its hump is the model’s analogue of their cyclical index, driven by task destruction and creation rather than by the business cycle. Closest on the question of tempo, Adão, Beraja, and Pandalai-Nayar (2024) relate the speed of a technological transition to skill specificity: when the benefited activities require skills far from the rest of the economy, incumbents cannot reallocate and demand is met by the entry of new generations. In their model the economy converges to the endpoint the technology pins down, regardless of the path. Here the tempo of the shock races a fixed speed of reallocation, and the endpoint moves with it: the same technology, run slowly or quickly through the same economy, binds the reinstated work to different occupations. The calibration remains illustrative, in the discipline of Storck (2026): no causal claim is made from it.

2 The static foundation

We recall only what the dynamics require. The geometry — the disk, its coordinates, and the mapping of tasks and occupations into it — is constructed and validated in Storck and Andersson (2026); the model of technological change on it, and the calibration used

here, are from Storck (2026).

Tasks lie on the unit disk in polar coordinates $\mathbf{r} = (\xi, \chi)$, with ξ a domain orientation and χ a depth. The market price of skill is a directional field

$$\ln \Pi(\mathbf{r}) = m_0 + m_1 \cos \xi + m_2 \sin \xi + \chi (m_3 + m_4 \cos \xi + m_5 \sin \xi), \quad (1)$$

high in the socio-cognitive directions and low in the service and manual ones (Storck and Andersson 2026). Depth is what Adão, Beraja, and Pandalai-Nayar (2024) call skill specificity; the geometry gives it a direction, and the price field decides where specificity pays. A technology is a Gaussian field $\phi_K(\mathbf{r}) = A_K \exp(-\frac{1}{2}\|\mathbf{r} - \mathbf{p}_K\|^2/z_K^2)$, with maturity A_K , centre $\mathbf{p}_K$, and reach z_K . Capital bears a task only where it is the cheaper producer, so the operated share is

$$a(\mathbf{r}) = \left[1 + \exp\left(-\left(s_K \phi_K(\mathbf{r}) - R/\Pi(\mathbf{r})\right)/\tau\right) \right]^{-1}, \quad (2)$$

where τ is the width of the adoption margin, and the condition crosses first where the price Π is high: automation takes the dearest work. Each occupation o is a bundle $b_o(\mathbf{r})$ with centroid μ_o ; its displaced mass is $D_o = \int b_o a$, and the aggregate displacement is $\Delta\Gamma^D = \sum_o L_o D_o$.

Reinstatement creates new task mass $M = \gamma \Delta\Gamma^D$ on the gradient ring of the field, $\hat{g}(\mathbf{r}) = \|\nabla\phi_K\|/\int\|\nabla\phi_K\|$, which peaks at distance z_K from the centre. A seeded task survives as human work only where the operated share is low, through the survival gate $(1 - a)$: deep in the core, where capital dominates, the same seed is captured by capital. The surviving seed density is therefore $\hat{g}(1 - a)$, concentrated on the boundary crescent. An occupation can take a seeded task only where its capabilities match the task's requirements. The implemented readiness, eq. (attachment) of Storck (2026), is symmetric and local,

$$e_o(\mathbf{r}) = \exp\left[-\left(\delta_o(\mathbf{r}) + \lambda_{\text{over}} \bar{\delta}_o(\mathbf{r})\right)/\ell\right] \exp\left(-\|\mathbf{r} - \mu_o\|/\rho\right), \quad (3)$$

where $\delta_o = \sum_k v_k \max(q_k(\mathbf{r}) - q_{o,k}, 0)$ is the v -weighted capability deficit, $\bar{\delta}_o$ the corresponding surplus, ℓ the acquisition scale, ρ the attachment locality, and λ_{over} the over-qualification weight. The attachment capacity of a location is $C(\mathbf{r}) = \sum_o L_o e_o(\mathbf{r})$. This primitive is shared verbatim between the two papers and between the static and dynamic layers of the implementation.

All parameters of this foundation are inherited unchanged from the economy table of Storck (2026): $R = 18$, $\tau = 0.08$, $\beta = 0.5$ (congestion), $\gamma = 0.5$, $\ell = 0.133$, $\rho = 0.5$, $\lambda_{\text{over}} = 1$. The mobility scale ν (one standard deviation of baseline occupation value per logit unit) and the move cost c_m (the median move costs one ν) follow the same rules, evaluated here at the pre-shock baseline $A_K = 0$, the state the dynamics start from: $\nu = 11.8$ and $c_m = 23.0$. That table evaluates the same rules at the calibrated technology

and reports 11.6 and 22.6; the two per cent difference moves no result at its reported precision.

Two features of this foundation are load-bearing for what follows. First, displacement falls *within* occupations, on the interior margin $a(\mathbf{r})$, not between regions. Second, the survival gate places the surviving reinstatement on the low-priced side of the boundary: humans retain the work that is not worth automating, which in this price field is the cheaper, lower- Π work. The destination results below interact with this second feature, and we return to it in the discussion.

3 From equilibrium to motion

The static and dynamic models are not one model seen two ways; they are different models that answer different questions. The static model computes a state $L = F(L)$ without intervening time; its solver's intermediate steps are numerical artefacts and its damping is a convergence parameter, not a speed. The dynamic model writes laws of motion — differential equations in time — and lets the state move under them, with inertia and an actual elapse of time. That the dynamic state may approach a rest point is a consequence of the laws, not something computed, and in general it is never reached, because the target shifts while the technology matures.

System dynamics supplies the grammar: stocks change only through flows, and flows form loops. The geometry supplies the constitutive laws: every flow rate is a functional of the geometric fields. The technology maturity $A_K(t)$, the population $L_o(t)$, the unbound stock $U(\mathbf{r}, t)$, and the slow drift of occupational character are the stocks; of these, only U is new.

3.1 The unbound stock

In the static model the seeded mass M has three fates: a part sa is captured by capital, a part $s(1-a)\Phi$ binds to occupations, and a part $s(1-a)(1-\Phi)$ is unbound, where $s = M\hat{g}$ and Φ is the bound share. The unbound part performs no work and does not enter the density; it is created and discarded at each evaluation. The dynamic model gives it a stock,

$$\partial_t U(\mathbf{r}, t) = \dot{s}(\mathbf{r}, t) - \sum_o \iota_o(\mathbf{r}, t), \quad \dot{s}(\mathbf{r}, t) = \gamma [\partial_t \Delta \Gamma^D]_+ \hat{g}(\mathbf{r}) (1 - a(\mathbf{r}, t)), \quad (4)$$

where $\dot{s}$ is the seeding rate and ι_o the binding flow to occupation o (§3.3). The seeding follows the *rate* of displacement: new tasks are created while the technology matures, and seeding ceases once displacement plateaus. The unbound stock is the only structural addition; everything else is the static model read as a rate.

A point of bookkeeping matters here. The dynamics do not impose global conservation of task mass, and should not: seeding is a genuine source (new tasks are created). What is conserved is the population, $\sum_o \dot{L}_o = 0$, and the disposition of created task mass among its three fates. The stock U closes the seeding ledger — it gives the unbound fate a carrier — not a spurious global conservation.

3.2 Population and maturation

The population moves toward the static reallocation target, but from the *current* distribution rather than a frozen baseline:

$$\dot{L}_o = \frac{1}{\theta_L} (T_o(L, t) - L_o), \quad T_o(L, t) = \sum_{o'} L_{o'}(t) P(o | o'; W(L, t)), \quad (5)$$

with P a row-stochastic origin–destination logit, so $\sum_o \dot{L}_o = 0$ exactly. The timescale θ_L is a calibrated speed in interpretable units; it replaces the static solver’s damping, inheriting nothing from it. First order is deliberate: it yields lags and gaps but no overshoot.

The technology matures as a logistic diffusion anchored in calendar time. With t in years and a shock window T_{shock} , the slope is set so that A_K rises from five to ninety-five per cent of its mature value over T_{shock} years, centred at $T_{\text{shock}}/2$:

$$A_K(t) = \frac{\bar{A}_K}{1 + \exp\left(- (5.88/T_{\text{shock}}) (t - T_{\text{shock}}/2)\right)}. \quad (6)$$

Anchoring the shock in calendar time makes θ_L and the binding timescale below calendar quantities, so that the outcome is governed by the *ratio* of the shock tempo T_{shock} to the redistribution tempo. We use $T_{\text{shock}} = 5$ years throughout as an illustrative diffusion window. Only the ratio of T_{shock} to the redistribution tempo enters the results: all rates in the model scale as $1/\theta$ or are set by the diffusion window, so the choice of window is a normalisation of the time axis, and the sweep over θ at fixed T_{shock} traverses the ratio.

3.3 Binding: match allocates, size limits the rate

The binding flow ι_o rests on a separation that proved load-bearing: *where* the unbound mass goes is set by match, *how fast* it binds is set by size. At each location the mass is attracted to the best match through a sharpened share, and each occupation binds up to a size-limited capacity:

$$t_o(\mathbf{r}) = U(\mathbf{r}) \frac{e_o(\mathbf{r})^{\beta_m}}{\sum_{o'} e_{o'}(\mathbf{r})^{\beta_m}}, \quad (7)$$

$$c_o = M_o/\theta_{\text{abs}}, \quad M_o = b_o^{\text{orig}} + r_o, \quad \iota_o(\mathbf{r}) = t_o(\mathbf{r}) \min\left(1, \frac{c_o}{\int t_o d\mathbf{r}}\right). \quad (8)$$

The claim t_o distributes the unbound mass by sharpened match: the readiness e_o is the shared attachment primitive of eq. (3), and the exponent $\beta_m > 1$ lets the best match dominate the claim at each location. The capacity c_o scales with the occupation’s task mass M_o , which grows endogenously with the reinstated mass r_o it has already bound. What a saturated occupation cannot staff this step — the fraction $1 - \min(1, c_o/f t_o)$ — returns to U and is offered, at the next step, to the next-best match. Large occupations therefore bind faster in absolute terms but not faster relative to their size, and the residue cascades down the match ranking.

The same primitive sets the static attachment capacity C and allocates the unbound mass here; the static and dynamic layers of the implementation share it exactly. Its symmetry carries the economics: being able to do a task and becoming its home are different questions, and penalising over-qualification alongside the deficit keeps an occupation far above a task’s level from drawing the task in as its own work. Storck (2026) states the form and its one-sided limit ($\lambda_{\text{over}} = 0, \rho \rightarrow \infty$). What the dynamic layer adds is only the sharpening β_m and the size-limited rate.

The form was chosen against an alternative that conflates the two roles. A binding rate proportional to size times match, $\iota_o \propto M_o e_o$, lets size *multiply* match: in that form the absolute absorption correlates with occupation size at +0.99 and with the unconstrained claim at +0.07, at the reference tempo — size decides the destination and match governs only the relative growth. The match-allocated form with a size cap inverts this: absorption tracks the claim at +0.73, and the size correlation falls to +0.13 at the reference tempo and to +0.04 in the gradual regime, with size demoted from setting *where* the work goes to *how fast* it binds, which is its proper role. The sharpness β_m and the absorption timescale θ_{abs} are calibration handles, not worldviews; their economic content — whether absorptive capacity scales linearly or sub-linearly with size, and how sharply the best match wins — is settled by a sweep, and the central result below is itself such a sweep. Their economic interpretation is direct. The sharpness β_m sets how strongly the best match dominates the local claim: $\beta_m = 1$ shares the mass in proportion to fit, $\beta_m \rightarrow \infty$ assigns each location strictly to its best match; its calibration target is the observed concentration of new work across occupations. The absorption timescale θ_{abs} is the time an occupation needs to absorb its own task mass in new work, an onboarding capacity per unit size; its target is the observed pace at which occupations take on new task content. The runs fix $\beta_m = 3$ and sweep the tempo; Section 4.2 reports the result’s sensitivity to β_m over 1 to 8 and shows that the destination result is carried by θ_{abs} alone.

3.4 Limit allocations

The two roles that the binding law separates become exact in the two limits of the absorption tempo, and the destination result of Section 4 is the path between them.

Proposition 1 (fast absorption) *Fix the occupation set. As $\theta_{\text{abs}}/T_{\text{shock}} \rightarrow 0$ no capacity ever binds, and the final allocation of reinstated mass is*

$$R_o = \int w_o(\mathbf{r}) s(\mathbf{r}) d\mathbf{r}, \quad w_o(\mathbf{r}) = \frac{e_o(\mathbf{r})^{\beta_m}}{\sum_{o'} e_{o'}(\mathbf{r})^{\beta_m}}, \quad (9)$$

where s is the cumulative surviving seed density. As $\beta_m \rightarrow \infty$ the shares concentrate on the best match and the allocation converges to strict best-match assignment.

Proof. With slack capacities the absorbed fraction is one for every occupation at every step, so each step's seeding is bound in full at the claim shares of eq. (7). Absent births the readiness fields are time-invariant, so the shares w_o are as well, and summing over steps gives $R_o = \int w_o s$. The second statement is the pointwise limit of the sharpened shares. $\square$

The statement is exact at the level of the discrete law: at $\theta_{\text{abs}} = 0.05$ the realised allocation matches $\int w_o s$ to machine precision (relative L_1 error below 10^{-15}).

Proposition 2 (slow absorption) *As $\theta_{\text{abs}}/T_{\text{shock}} \rightarrow \infty$ the final allocation converges to the initial size shares, $R_o \rightarrow S M_o(0)/M_{\text{tot}}$, with S the total surviving seed mass.*

Proof. Every claim share is strictly positive, since the readiness is an exponential, so each occupation's desired intake is positive wherever unbound mass remains, while the capacities M_o/θ_{abs} shrink without bound; for θ_{abs} large enough every capacity binds throughout the drain. While they bind, absorption follows $\dot{r}_o = M_o(t)/\theta_{\text{abs}}$ with $M_o(t) = M_o(0) + r_o(t)$, whose solution $r_o(t) = M_o(0) (e^{t/\theta_{\text{abs}}} - 1)$ is a common exponential factor times the initial mass: endogenous task-mass growth preserves the proportions. The phases in which some capacity is slack — the extreme tails of the seeding pulse, and the end of the drain where the remaining unbound mass falls below the largest capacity — carry a mass share that vanishes as θ_{abs} grows. $\square$

While all capacities bind, the aggregate obeys $\dot{U}_{\text{tot}} = \dot{s}_{\text{tot}}(t) - M_{\text{tot}}/\theta_{\text{abs}}$, so after seeding ends the stock drains linearly and empties in $\theta_{\text{abs}} S/M_{\text{tot}}$ years. The bound is saturated only in the deep limit: the capacity utilisation — the mass absorbed per step over the aggregate bound $(\Delta t/\theta_{\text{abs}}) M_{\text{tot}}$, read at the peak of the mismatch — rises from 0.11 at $\theta_{\text{abs}} = 2$ to 0.56 at 15 and 0.99 at 120. The shortfall is the spatial locality of binding: claims are proportional to $e_o^{\beta_m}$, so the mass on the seeding crescent is in practice drained by the occupations near it, and the aggregate task mass overstates the absorbing mass at intermediate tempos. This is why the mismatch hump of Section 4.1 outlives the naive aggregate prediction.

4 Results

We run a single technology field, calibrated to task-level AI exposure as in Storck (2026), through the dynamic economy. The field matures over $T_{\text{shock}} = 5$ years; we vary the redistribution tempo θ (taking θ_L and θ_{abs} together) against this fixed shock. The economy parameters are the static model's, restated in Section 2; the dynamic layer adds the redistribution timescales θ_L and θ_{abs} (both 3 years at reference), the claim sharpness $\beta_m = 3$, the diffusion window $T_{\text{shock}} = 5$ years, and the step $\Delta t = 0.2$ years. Occupational birth is switched off in the runs of this section, so the occupation set is identical across regimes and the allocation comparison is well defined. In every run the population sum is conserved to numerical precision and the unbound stock drains below 10^{-8} ; both are asserted by the producing scripts.

4.1 The transitional mismatch

Figure 1 shows the shock and the mismatch it creates. The left panel is the logistic diffusion of (6), rising over the five-year window. The right panel is the unbound stock $U(t)$ for four redistribution tempos, $\theta \in \{1, 3, 8, 15\}$ years. When redistribution is faster than the shock, the mismatch is barely visible: the market binds the displaced work as fast as it is seeded. As redistribution slows, the mismatch hump grows — by a factor of eighty from the fastest to the slowest tempo shown — and peaks later, moving from $t \approx 2.6$ to $t \approx 3.6$ years within the adoption window. In every case it drains to zero once diffusion saturates and binding catches up. The shape is that of the mismatch index Şahin et al. (2014) measure over the Great Recession, which likewise rises and drains with no structural shift; the driver here is task destruction and creation, not the business cycle. There is no permanent mismatch in the model: all created work is eventually bound. The transitional cost lives entirely in the path — in the height and persistence of the hump — not in the endpoint, which is exactly what the static model, seeing only the endpoint, cannot register.

4.2 The destination depends on the tempo

The hump measures the cost of the transition; the same ratio also determines its outcome. Figure 2 shows where the reinstated work is bound in two regimes, run through the same technology and the same field. In the gradual regime ($\theta = 1$ year against the five-year shock), the best-matched occupations keep pace and absorb the work: the gainers are the high-skill specialists whose capabilities sit at the frontier of the field. In the congested regime ($\theta = 15$ years), the small best-matched occupations cannot bind fast enough — their size-limited capacity $c_o = M_o/\theta$ is small, the mass accumulates in U , and it is taken instead by the large, lower-skilled occupations that can absorb it quickly. The same work

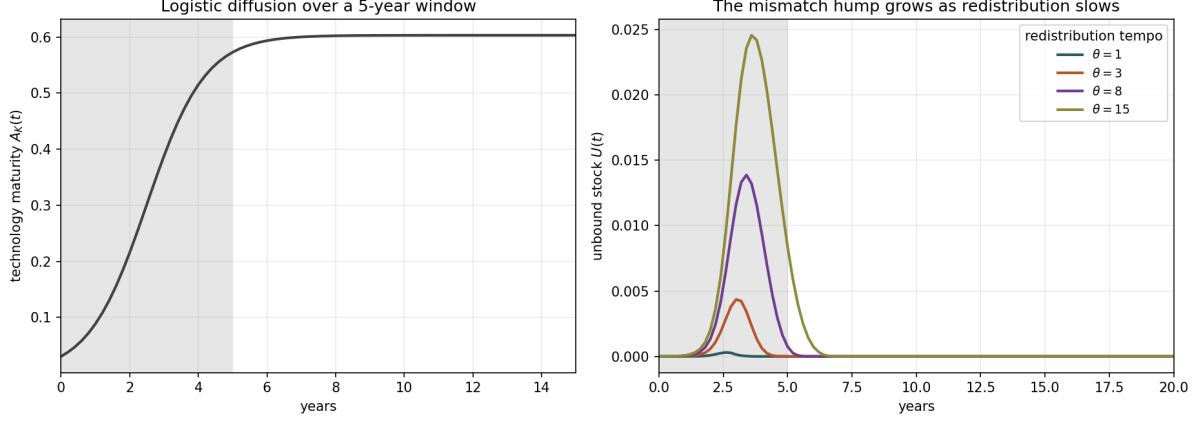


Figure 1: Calendar-time dynamics. Left: the technology matures as a logistic diffusion over a five-year window. Right: the unbound stock $U(t)$, the transitional mismatch between destruction and creation, for four redistribution tempos $\theta \in \{1, 3, 8, 15\}$ years against the fixed five-year shock. The mismatch hump grows and lingers as redistribution slows relative to the shock, but always drains to zero: the transitional cost is in the path, not the endpoint.

is downgraded. The leading gainers make the point concretely: in the gradual regime they are preventive-medicine physicians, physicists, and natural-sciences managers; in the congested regime, fast-food and counter workers, waiters, and dishwashers.

The shift is not marginal. The final allocations of reinstated mass in the two regimes correlate at +0.37 (Pearson, unweighted across the 849 pre-existing occupations; +0.31 by rank): substantially different destinations from one technology and one field, differing only in the tempo of the shock relative to the market’s capacity to respond. Weighting by baseline employment raises the correlation to +0.68, because the largest occupations gain in both regimes; the divergence is concentrated in whether specialist best matches or large absorbers take the frontier work, which is the margin the result is about. This is the destination counterpart of the transitional hump, and it is a statement the equilibrium framework cannot make. The fixed point sees the endpoint of a frictionless reallocation; it cannot see that the endpoint the economy actually reaches depends on how fast it was forced to move.

The mechanism reads off the per-occupation evidence. Define an occupation’s unconstrained claim as the seed-weighted share the claim law (7) would allocate with no size cap, and its size as the pre-shock task mass that caps the rate. In the gradual regime absorption tracks the claim — Pearson +0.83, rank +0.92 — and is uncorrelated with size (+0.04); the top claim quartile takes 49 percent of the reinstated mass. In the congested regime the claim decouples (+0.37, rank +0.21) and size takes over (rank +0.95); the top size quartile takes 57 percent of the mass and the bottom quartile is squeezed from 22 to 4 percent. All rankings are in absolute mass; the relative-change measure is unbounded for small occupations and is not used.

Reinstated task mass gained per occupation (final allocations correlate at +0.37)
 Gradual transition ($\theta = 1 < T_{shock} = 5$) Congested transition ($\theta = 15 > T_{shock}$)

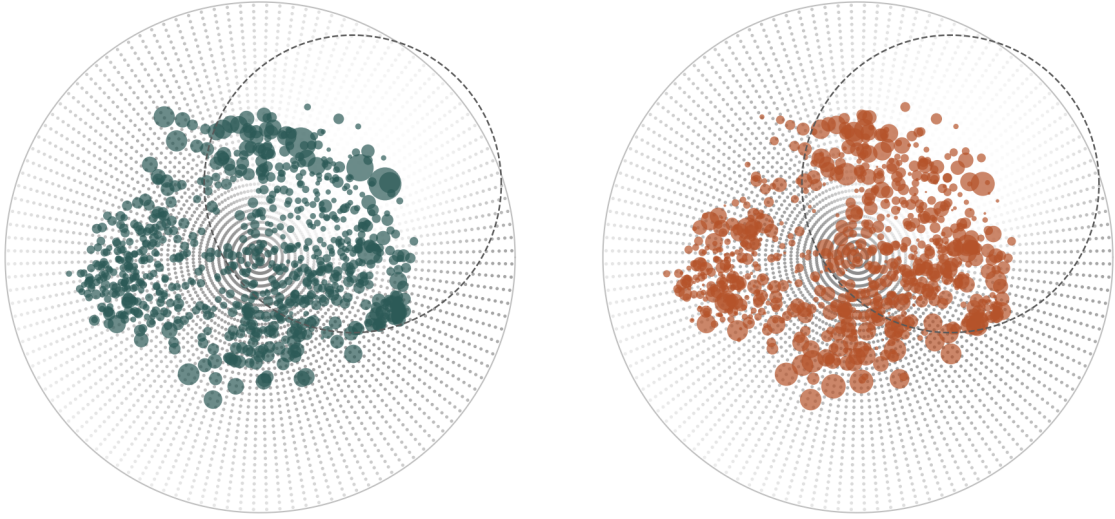


Figure 2: The destination of reinstated work depends on the tempo ratio. Both panels show the reinstated task mass gained by each occupation, over the survival-gated seeding field (grey) with the technology field dashed, for the same five-year shock. Left, a gradual transition: the best matches keep pace and high-skill specialists at the frontier absorb the work. Right, a congested transition: small best-matched occupations cannot keep up and the work is forced onto large, lower-skilled occupations. The two final allocations correlate at +0.37.

Three robustness checks bound the claim. Sweeping the sharpness β_m from 1 to 8 moves the cross-regime correlation from +0.26 to +0.61: the divergence of destinations survives the full range, while its point size depends on the handle. And decomposing the tempo into its two timescales shows which one binds: slowing only absorption ($\theta_L = 1$, $\theta_{abs} = 15$) reproduces the congested allocation (correlation +0.39 with the gradual run), while slowing only mobility ($\theta_L = 15$, $\theta_{abs} = 1$) leaves the destination essentially unchanged (+1.00). The forced downgrading is a statement about how fast occupations can take on new work, not about how fast workers move.

The third check varies the survival gate itself, since the low-priced character of surviving seed is inherited from it. Both regimes are rerun with the gate off, and with a comparative-advantage gate: a human-productivity field $h(\mathbf{r}) = \exp[\lambda_h(1 - \hat{\phi}_K(\mathbf{r}))]$, with $\hat{\phi}_K$ the unit-amplitude technology shape, so labour is more productive where the machine is weak and capital operates where $s_K\phi_K > hR/\Pi$; λ_h is swept up to 2, and $\lambda_h = 0$ recovers the uniform- R gate. The divergence is intact in every variant — the cross-regime correlation stays between +0.35 and +0.46, and the congested size rank correlation between +0.91 and +0.96. What the gate moves is the gainers' price level, not the tempo dependence: without the gate the capital-dominated core seeds, and the congested absorbers become large health and administrative occupations rather than food service; the comparative-advantage gate raises the mass-weighted gainer price with λ_h . The downgrading — congested gainers

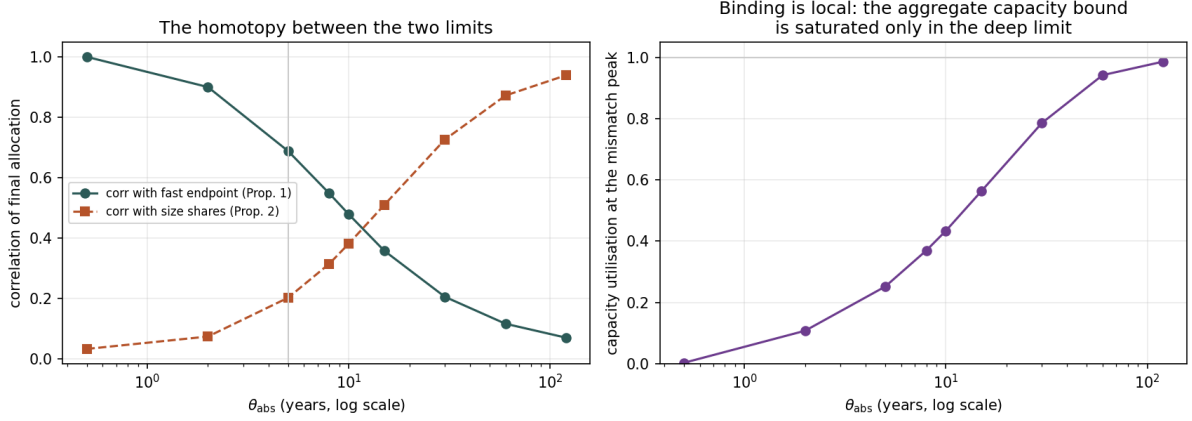


Figure 3: The destination as a homotopy between proved limits. Left: the correlation of the final allocation with the fast-limit claim allocation (Proposition 1) and with the slow-limit size shares (Proposition 2), across the absorption tempo at fixed mobility; the vertical line marks the shock window. Right: the capacity utilisation at the mismatch peak, the mass absorbed per step over the aggregate bound; the shortfall at intermediate tempos is the spatial locality of binding, and Proposition 2 is the regime where it reaches one.

priced below gradual gainers — persists throughout.

4.3 The homotopy between the limits

The tempo result is the path between the two proved endpoints of Section 3.4. Figure 3 traces the final allocation across θ_{abs} from 0.5 to 120 years against the fixed five-year shock, with mobility held fast ($\theta_L = 1$; the decomposition above shows the destination is carried by θ_{abs}). Its correlation with the fast-limit allocation of Proposition 1 falls monotonically from +1.00 to +0.07, and its correlation with the size shares of Proposition 2 rises monotonically from +0.03 to +0.94 (+0.996 by rank) over the same sweep. The crossover, where the two associations are equal, sits between $\theta_{\text{abs}} = 10$ and 15: two to three shock windows. The simulation’s remaining role is to locate this crossover between two allocations that are derived, not simulated.

5 The general-equilibrium price feedback

Both the static regime and the dynamic transition above hold the price field $\Pi(r)$ fixed: the shock relocates labour and re-sorts it, congestion adjusts the marginal value of crowded destinations, but the per-unit price of capability content does not re-solve. This is a short-run reading. The price field is an equilibrium object, derived as a capability assignment in Storck (2026), Section 3.3, and an equilibrium that does not equilibrate discards its own content. Acemoglu and Restrepo (2026) make the point sharply: in their task model the

base wage *adjusts* to automation (their propagation matrix Θ), and that adjustment is the mechanism. Our fixed-price layers hold even more fixed than they do — both the rent wedge and the base price — so to “derive the prices” in their sense we must let the base price move. This section closes that loop.

The closure: level from data, change from the assignment. We keep the level from the estimated field — the empirical anchor — and take the *change* from the assignment, exactly as Acemoglu and Restrepo (2026) take the wage level as given and compute its response. Their wage law $w_g \propto (\Gamma_g/\ell_g)^{1/\lambda}$ ties the wage to tasks per worker; its spatial form is

$$\Delta \ln \Pi(r) = \frac{1}{\sigma} \left[\ln(1 - a(r)) - \ln \frac{n_L(r)}{n_L^0(r)} \right], \quad (10)$$

applied to the baseline, $\Pi(r) = \Pi_0(r) e^{\Delta \ln \Pi(r)}$. Automation is a *demand* shock to labour: where capital takes the task (a high), the task leaves the human set and the wage falls; and displaced labour crowding the remaining work ($n_L > n_L^0$) lowers it further. Both terms are stabilising. The sign is not innocuous: the opposite reading — treating automation as a *supply* shock that makes scarce human work dearer — inverts equation (10) and is destabilising, since a dearer wage opens the takeover gate. Run as a single pass at $\sigma = 3$, the scarcity reading collapses the labour share to 0.33 (capital chases the work it has just made dear), while the demand reading lifts it to 0.83; the fixed-price value 0.58 sits between. Labour shares in this section hold the allocation at its baseline, which isolates the price channel; the headline shares of Storck (2026) re-solve the allocation as well and are correspondingly higher — 0.626 at fixed prices, 0.78 at the fixed point below. The headline is not robust to the sign, which therefore must be chosen on economic grounds. Acemoglu and Restrepo (2026) choose ours: automation dissipates rents and lowers wages, it does not raise them.

The static fixed point. Iterating price and allocation to mutual consistency converges, and the convergence follows from a condition that can be evaluated at the calibration. Differentiating the gate (2) gives $\partial \ln(1 - a)/\partial \ln \Pi = -a(R/\Pi)/\tau$: a lower price closes the gate, which raises the price back. The slope of the map (10) is therefore bounded by

$$K \leq \frac{1}{\sigma} \left(\sup_{\mathbf{r}} \frac{a(\mathbf{r}) R}{\Pi(\mathbf{r}) \tau} + \varepsilon_n \right), \quad (11)$$

where ε_n is the crowding elasticity of the re-sorted density, and damped iteration with weight d converges when the stabilising slope satisfies $K < 2/d - 1$; at $d = \frac{1}{2}$ the threshold is 3. Evaluated at the converged point, the gate term peaks at 1.64 and the spectral radius of the full map, crowding included, is 1.59 by power iteration: the condition holds with a factor-two margin, and the iteration converges in nine damped steps. The slope scales as

$1/\sigma$ ($\sigma K = 4.77$ at $\sigma = 0.5$ against 4.78 at $\sigma = 3$), which places the failure boundary at $d = \frac{1}{2}$ near $\sigma = 1.6$; at $\sigma = 1$ the iteration indeed fails at $d = \frac{1}{2}$ and converges at $d = \frac{1}{4}$. The fixed point itself does not depend on the damping; only the path to it does.

The self-consistent state is an interior point, not the one-pass extreme: the labour share settles at 0.73 (against 0.58 at fixed prices), the price level falls 26 percent, and displacement is cut by roughly 44 percent (D_o from 0.28 to 0.16). The wage adjustment absorbs most of the shock: human work becomes about a quarter cheaper, capital adopts far less, and the labour share stays high — but the harm now sits in the wage *level*, not in employment, which is precisely the rent- and wage-dissipation of Acemoglu and Restrepo (2026).

The dynamic close, and statics meeting dynamics. Wired into the transition — the price responding to the current adoption and allocation at every step, equation (10) with a one-step lag — the feedback shapes the whole path, not just the end. The wage level falls monotonically as adoption rises, from -0.09 early to a plateau at -0.29 , and the displacement wave is markedly gentler: the unbound-mass (mismatch) peak is cut by about 63 percent (from 0.44 to 0.16 percent of the workforce) at the same timing, and the reinstated stock ends lower, because the falling wage deters the automation that would have produced the mismatch. Crucially, the transition lands where the static fixed point says it should: the dynamic end-state price level (-0.285) and labour share (0.739) reproduce the static fixed point ($-0.26, 0.73$) to within rounding. Two independent constructions — static fixed-point iteration and dynamic time-stepping — reach the same equilibrium, so the price feedback is a property of the model rather than of either method, and the static and dynamic accounts cohere.

What this does and does not establish. The feedback is first-order: at fixed prices both layers overstate the displacement and understate the labour share, and they overstate the violence of the transition. The stabilising wage adjustment is the shock absorber, and it operates throughout the transition, not only at its end. The quantitative point estimate depends on the assignment elasticity σ and on the reduced-form closure — equation (10) is the AR wage law applied region-wise, not a full re-solve of the assignment programme of Storck (2026) — and the AI field remains one illustrative instance of a technology field, not a measurement. Across σ from 5 to 0.5 the fixed-point lift of the labour share runs from $+0.11$ to a peak of $+0.20$ at $\sigma = 1$, and the damping of the mismatch hump rises from 46 to 89 per cent. We therefore read the result as establishing the *direction and order of magnitude* of the price feedback, and the mutual consistency of the static and dynamic closes, rather than a calibrated forecast of the labour share.

6 Discussion

The result is that the tempo of a technology shock, relative to the speed at which the labour market reallocates, is not merely a determinant of how painful the transition is but of its outcome. A gradual diffusion lets work find the occupations whose capabilities match it; a shock that outruns adjustment forces a downgrading that persists in the final allocation. The same field, the same displaced mass, and the same seeded reinstatement yield different societies depending on the pace. This is a genuinely dynamic statement, and it is the kind of statement the task-based equilibrium framework is not built to make: its rest point is the destination of a frictionless economy, and the friction here is time itself.

Two qualifications bound the claim, and both are inherited rather than introduced. The first concerns the survival gate. The static model places surviving reinstatement on the low-priced side of the boundary, because a seeded task is retained by labour where labour is the cheaper producer at a *uniform* cost R ; in Acemoglu and Restrepo (2019) reinstatement is instead new *complex* work in which labour has a comparative advantage, which requires the survival condition to turn on relative rather than absolute productivity. Section 4.2 runs exactly this variant, alongside the gate removed altogether, and the gate turns out to govern the *character* of the destination rather than its tempo dependence: the divergence and the size mechanism survive every variant, while the gainers' price level moves with the gate. What remains conditional on the survival-gate economics is therefore the high- versus low-skill reading of the congested absorbers — food service under the absolute gate, large health and administrative occupations without it — not the result that a fast shock forces the work onto whoever is large enough to take it.

The second qualification is the binding calibration. The allocation depends on the sharpness β_m and the absorption timescale θ_{abs} , and Section 4.2 reports both sensitivities: the divergence of destinations survives the sharpness sweep with a point size that moves with the handle, and it is carried by the absorption timescale alone. The remaining quantitative claim is the regime boundary — how slow absorption must be, relative to the shock, for downgrading to set in — which the tempo sweep exhibits rather than derives.

Several limitations remain. The realisation is illustrative, inheriting the calibration and discipline of Storck (2026): no causal claim is made from it. The unbound stock drains to zero, so the model has no permanent mismatch and no involuntary non-employment; the transitional cost is the hump, and a model with task decay or irreversible binding would carry a permanent imprint of the shock speed that this one does not. That extension is left for later work. New tasks enter as added weight on existing locations rather than as new points on the disk. And the relative-change measure of who gains is unbounded for small occupations, so the destination results should be read in absolute and spatial terms.

The downgrading of the fast regime has an empirical silhouette. Beaudry, Green, and Sand (2016) document a cascade after 2000: as demand for cognitive tasks fell,

high-educated workers moved down the occupational ladder and pushed the less educated further down and out. The direction is reversed here. In their account workers descend the ladder because demand for their tasks has fallen; in the fast regime the work itself descends the match ranking because the occupations that fit it best are too small to absorb it in time. The same outward pattern — work performed below the skill level that matches it — arises from different mechanisms, one demand-side, one absorption-side, and the cascading image is theirs.

7 Conclusion

Turning the spatial task-based model from an equilibrium into laws of motion requires one new object, a stock of unbound task mass, and yields two things the equilibrium cannot. The first is the transitional mismatch $U(t)$, the gap between destruction and creation as a transient that grows with the tempo of the shock and drains as the market catches up. The second, and the central result, is that the tempo of the shock relative to the speed of redistribution determines the destination of the reinstated work: a gradual shock matches it to high-skill specialists, a fast shock forces it onto large lower-skilled occupations. Both endpoints are proved as limits of the binding law, and the realised destination moves monotonically between them with the tempo. The same technology yields a different distribution of work depending only on how fast it arrives. The cost and the outcome of automation are, in this sense, properties of the path and not only of the endpoint.

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